

HEALTHCARE ANALYTICS DASHBOARD

Week 6 Task Report

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1. Introduction & Objective:

This report documents the Week 6 task: building an end-to-end Healthcare Analytics dashboard in Power BI on top of a hospital admissions and billing dataset. The work covers data modelling, DAX measure development, visual layout planning, and final dashboard assembly.

The goal of the dashboard is to give hospital administrators and finance teams a single view of patient volumes, billing performance and average length of stay, filterable by insurance provider, admission type, and date range. A readmission-rate measure is also built into the model for further analysis.

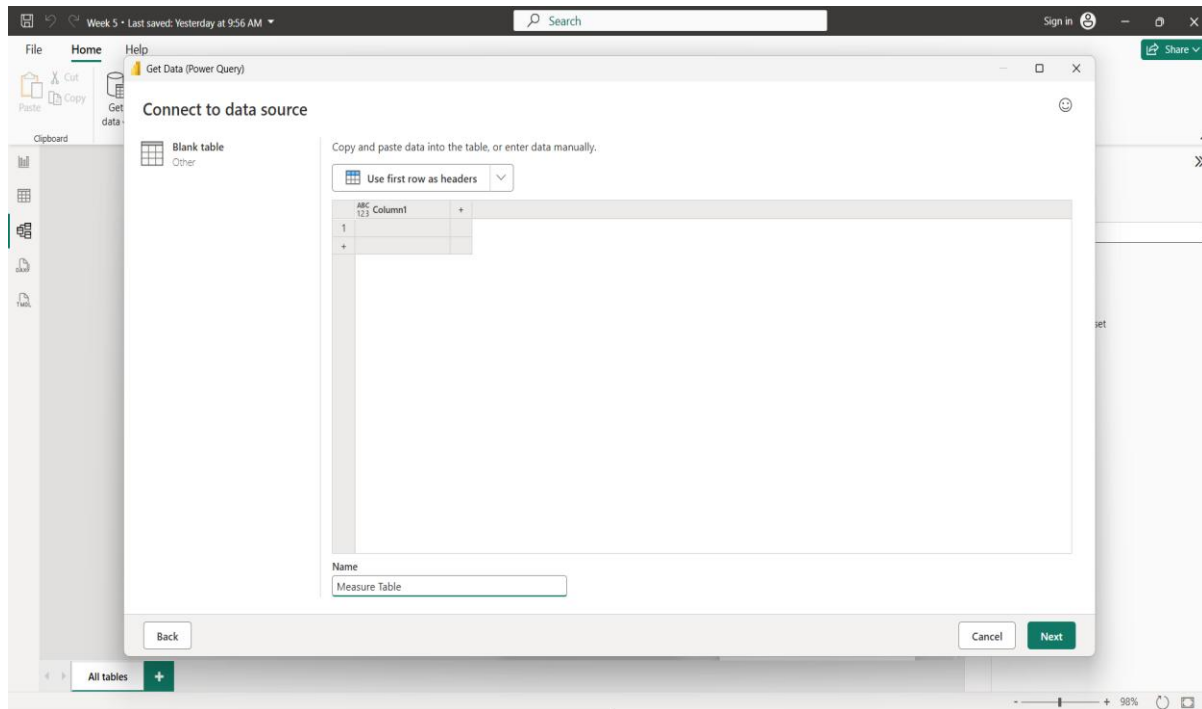
1.1 Dataset Overview:

The source data (healthcare_dataset) contains one row per billing record, with the following key attributes:

- Patient details: Name, Age, Age Group, Gender, Blood Type
- Admission details: Admission ID, Admission Type, Date of Admission, Discharge Date, Doctor, Hospital, Room Number
- Clinical details: Medical Condition, Medication, Test Results
- Financial details: Billing Amount, Insurance Provider

2. Data Model Setup:

Before writing any DAX, a dedicated Measure Table was created to keep all calculated measures separate from the raw data tables. This was done using Power BI's "Enter Data" (Get Data → Blank table) option, giving a clean home for every measure created in Section 3.



The final model is a star schema with a central Billing fact table connected to Dim_Patient and Dim_Admission dimension tables, plus the original flat healthcare_dataset kept for reference and the standalone Measure Table:

Table	Role
Billing	Fact table — Admission ID, Billing Amount, Date of Admission, Discharge Date, Insurance Provider, Patient ID
Dim_Patient	Dimension — Age, Age Group, Blood Type, Gender, Name, Patient ID
Dim_Admission	Dimension — Admission ID, Admission Type, Doctor, Hospital, Medical Condition, Medication, Room Number, Test Results
Measure Table	Houses all calculated DAX measures (no data columns)
healthcare_dataset	Original flat source table, retained for validation

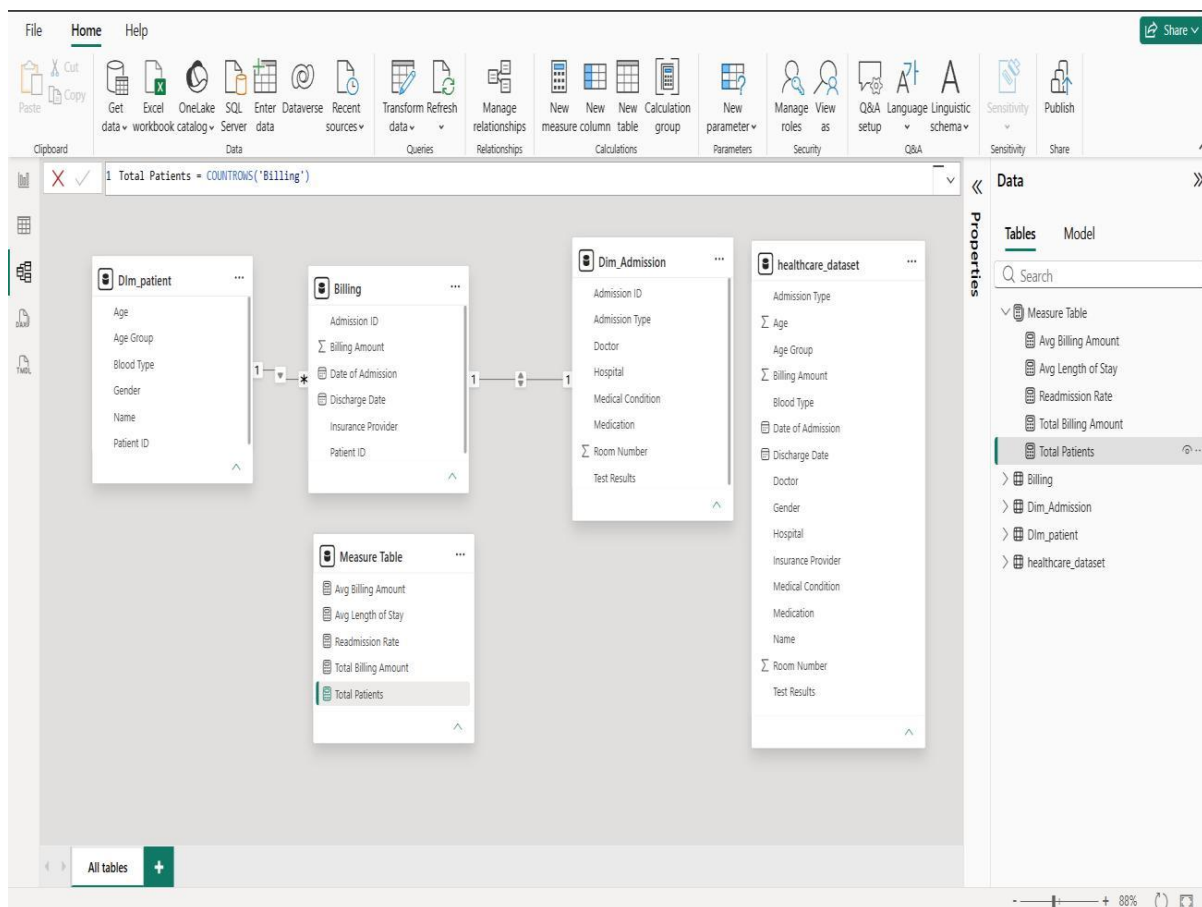
3. DAX Measures Development:

Five core measures were planned and then implemented against the model. Each measure was added to the Measure Table one at a time and verified in the Model view before moving to the next.

3.1 Total Patients:

Counts the number of billing/admission records, used as the base patient-volume KPI.

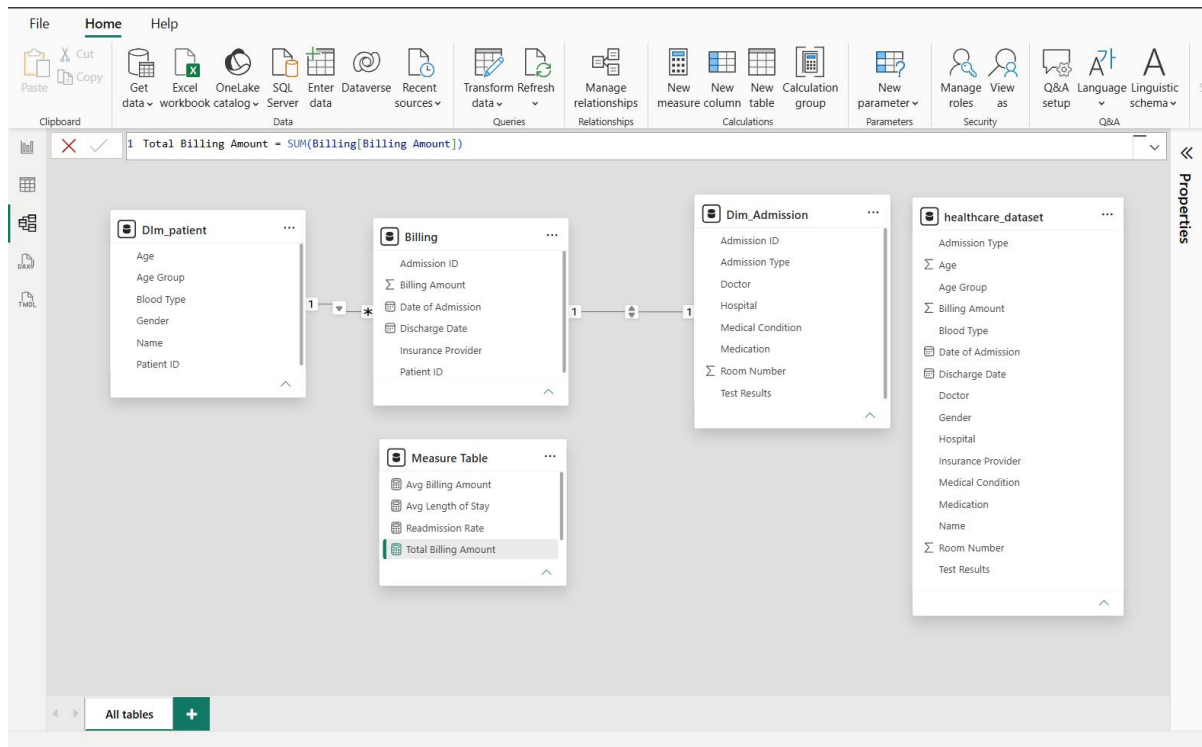
Total Patients = COUNTROWS(Billing)



3.2 Total Billing Amount:

Sums the billing amount across all records to give total healthcare revenue.

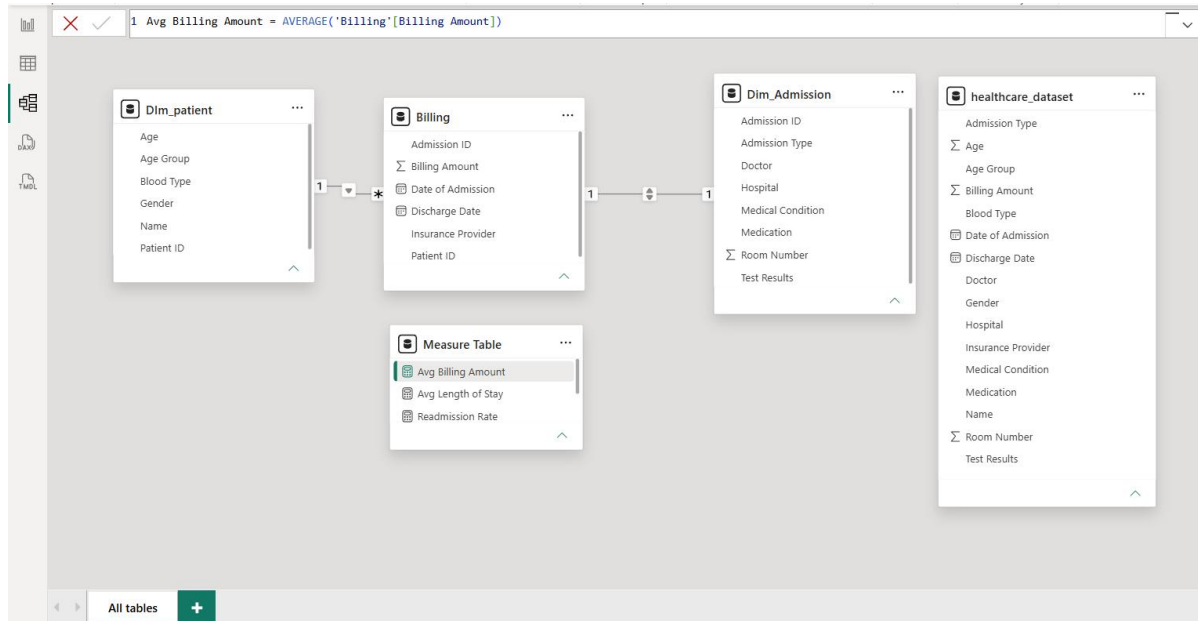
Total Billing Amount = SUM(Billing[Billing Amount])



3.3 Average Billing Amount per Patient:

Gives the average cost per billing record, useful for cost-per-patient benchmarking.

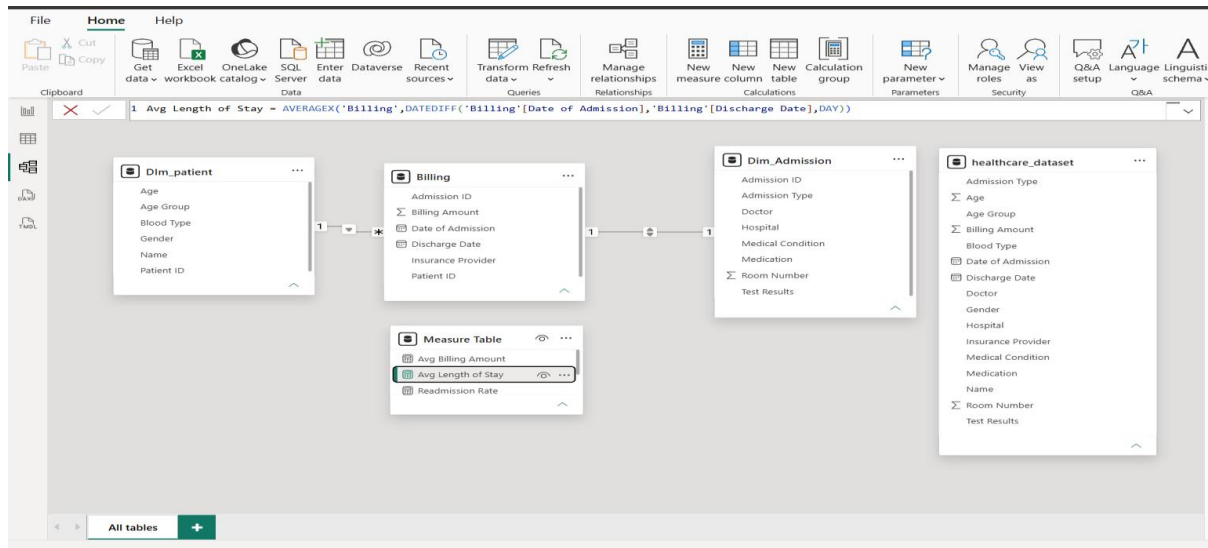
Avg Billing Amount = AVERAGE(Billing[Billing Amount])



3.4 Average Length of Stay (Days):

Calculates the average number of days between admission and discharge for each record using DATEDIFF inside AVERAGEX.

```
Avg Length of Stay =  
AVERAGEX(  
    Billing,  
    DATEDIFF(Billing[Date of Admission], Billing[Discharge Date], DAY)  
)
```



3.5 Readmission Rate:

Identifies patients who appear more than once in the Billing table (repeat admissions) and expresses them as a share of total patients, using SUMMARIZE + FILTER to build a per-patient admission count, then DIVIDE for a safe ratio.

```
Readmission Rate =  
VAR TotalCount = [Total Patients]  
VAR RepeatPatients =  
    COUNTROWS(  
        FILTER(  
            SUMMARIZE(  
                Billing,  
                Billing[Patient ID],  
                "Admission Count", COUNTROWS(Billing)  
            ),  
            [Admission Count] > 1  
        )  
    )  
RETURN  
    DIVIDE(RepeatPatients, TotalCount, 0)
```

The screenshot displays the Microsoft Power BI Desktop interface. At the top, the ribbon shows the 'Home' tab with various toolbars. The main workspace contains a DAX formula bar with the following code:

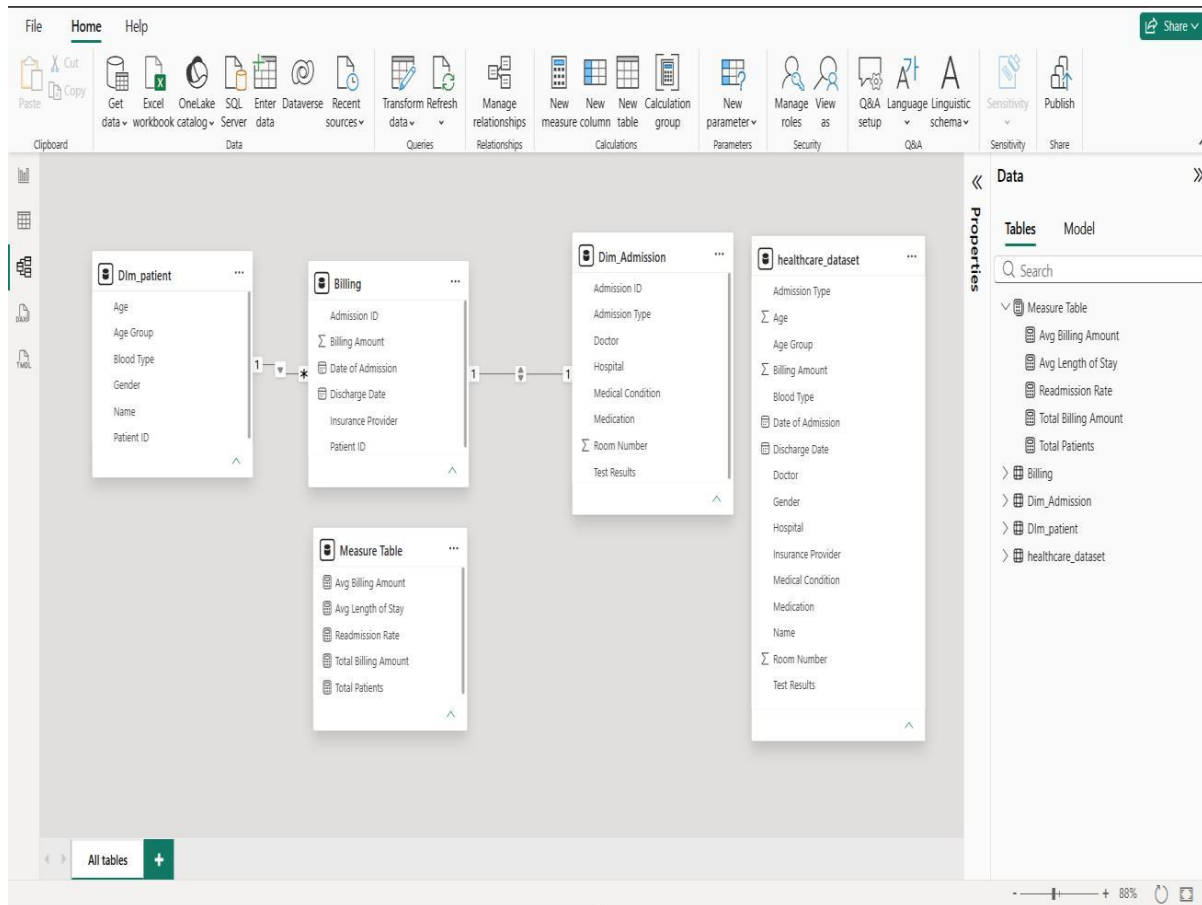
```
1 Readmission Rate = VAR TotalCount = [Total Patients]  
2 VAR RepeatPatients = COUNTROWS(FILTER(SUMMARIZE('Billing','Billing'[Patient ID],"Admission Count",COUNTROWS('Billing')),[Admission Count] > 1))  
3 RETURN DIVIDE(RepeatPatients,TotalCount,0)
```

Below the formula bar is a data model diagram showing four tables: Dim_patient, Billing, Dim_Admission, and healthcare_dataset. The Dim_patient table is connected to the Billing table with a one-to-many relationship. The Billing table is connected to the Dim_Admission table with a one-to-many relationship. The healthcare_dataset table is also connected to the Dim_Admission table. A 'Measure Table' is shown in the center, containing the following measures: Avg Billing Amount, Avg Length of Stay, Readmission Rate (highlighted), Total Billing Amount, and Total Patients.

On the right side, the 'Properties' pane is open, showing the 'Data' tab. The 'Tables' section lists the following tables: Dim_patient, Billing, Dim_Admission, and healthcare_dataset. The 'Model' section lists the following measures: Avg Billing Amount, Avg Length of Stay, Readmission Rate, Total Billing Amount, and Total Patients.

3.6 All Measures:

With all five measures in place, the Measure Table holds Avg Billing Amount, Avg Length of Stay, Readmission Rate, Total Billing Amount and Total Patients.



4. Visual Layout & Dashboard Design:

With the measures in place, the dashboard layout was planned before building, so every visual has a clear purpose and data-field mapping. The plan below was used as the build checklist for Section 5.

4.1 Top KPI Summary Cards:

- Total Patients
- Total Billing Amount
- Average Billing Amount
- Average Length of Stay

4.2 Demographics & Patient Distribution:

Matrix visual — Rows: Age Group | Columns: Medical Condition | Values: Total Patients, Average Billing Amount.

Donut chart — Legend: Age Group | Values: Total Patients.

4.3 Medical Conditions & Admission Volumes:

Stacked bar chart — Y-axis: Medical Condition | X-axis: Total Patients | Legend: Admission Type.

4.4 Admission Volume Trends & Hospital Analytics:

Line chart — X-axis: Date of Admission (Year, Month) | Y-axis: Total Patients, Total Billing Amount.

Column chart — X-axis: Insurance Provider | Y-axis: Total Billing Amount.

4.5 Slicers & Dynamic Cross-Filtering:

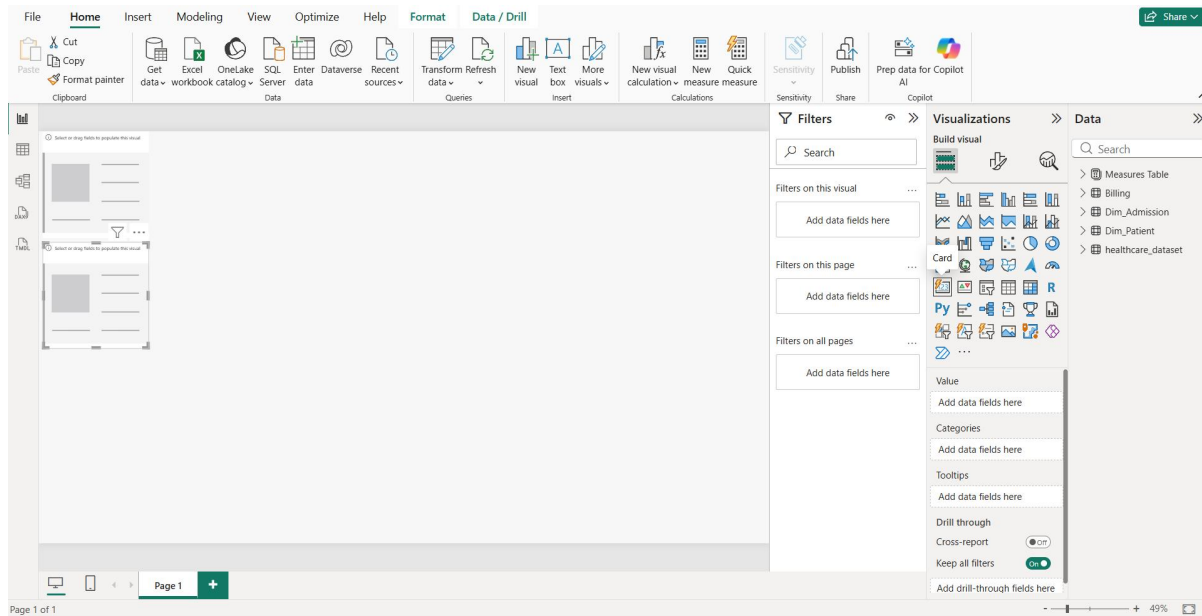
- Slicer 1 (List): Insurance Provider
- Slicer 2 (List): Admission Type
- Slicer 3 (Date Range): Date of Admission

5. Report Building Process:

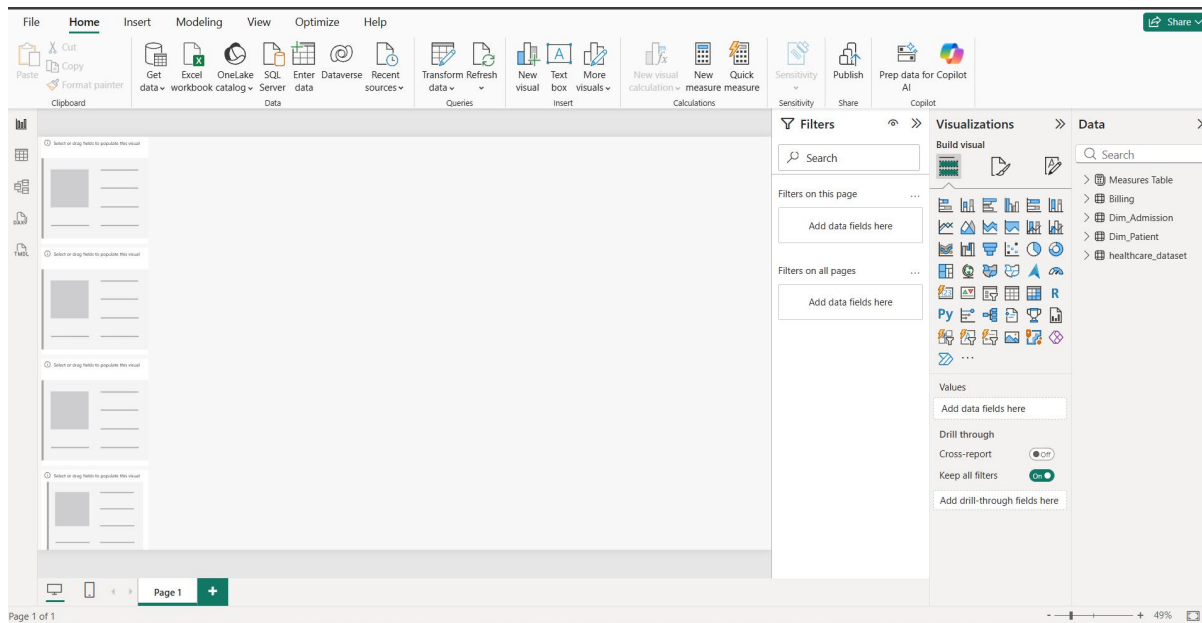
The visuals from the layout plan were added to the report canvas one at a time. Empty placeholders were inserted first to fix the layout, and the Fields pane was then used to bind each visual to the measures and columns from the model.

5.1 Adding KPI Card Placeholders:

A Card visual was inserted from the Visualizations pane. Its Value well is still empty at this stage, so it shows the “Select or drag fields to populate this visual” prompt. The data pane lists the Measures Table, Billing, Dim_Admission, Dim_Patient and healthcare_dataset.

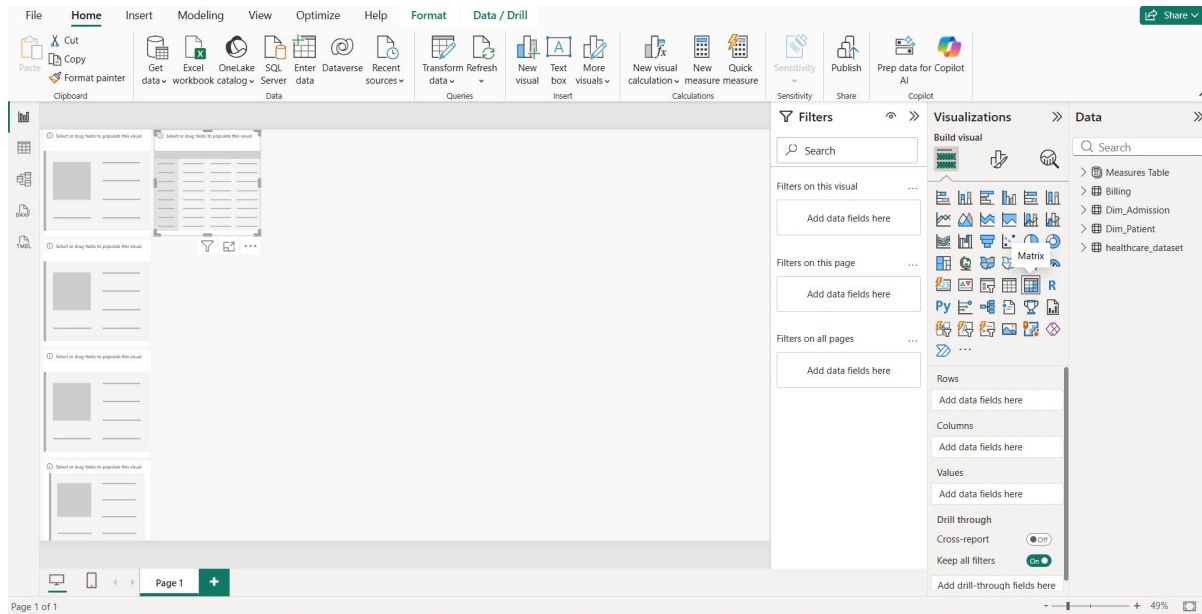


The card was then duplicated so that four cards are stacked in a column, one for each KPI in the plan.



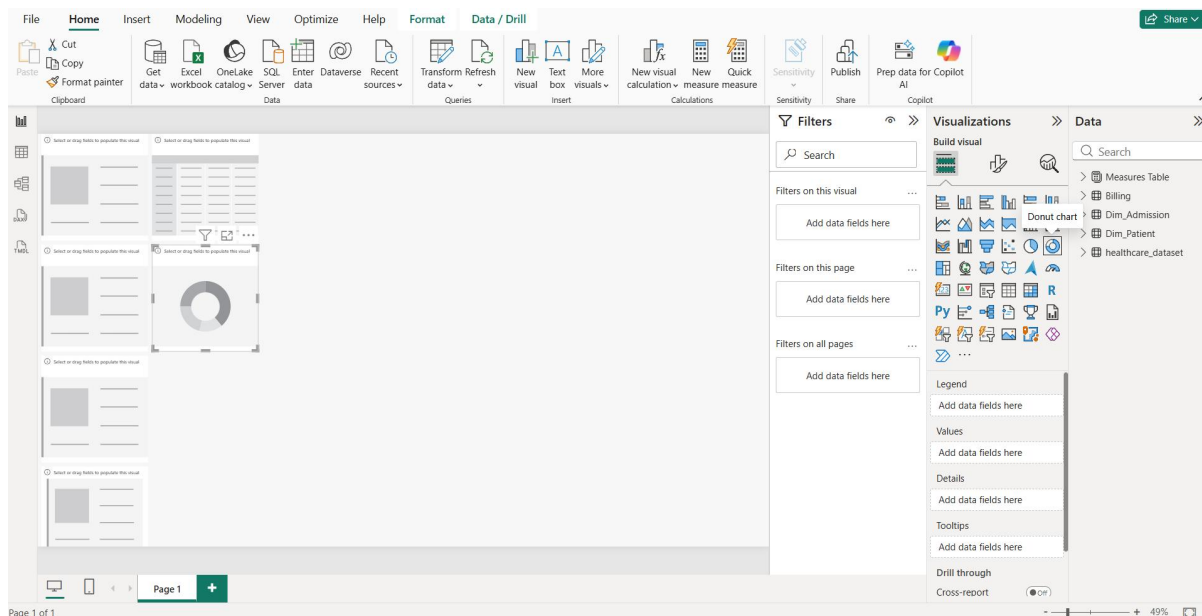
5.2 Adding the Matrix Visual:

A Matrix visual was placed next to the cards. Its field wells (Rows, Columns and Values) are used for the Age Group by Medical Condition breakdown.



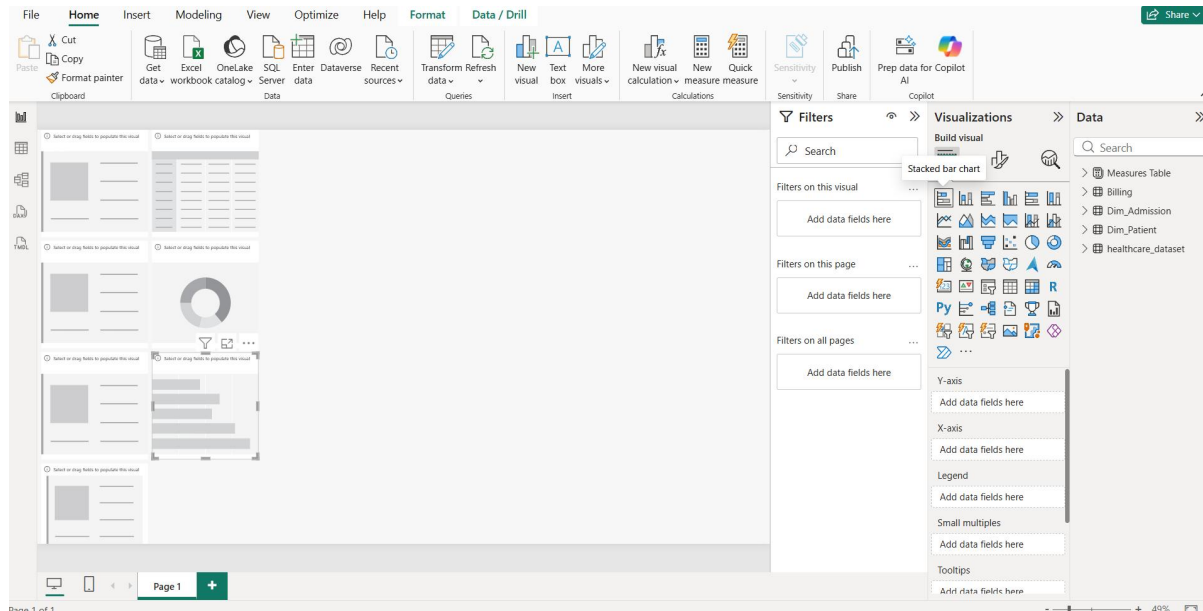
5.3 Adding the Donut Chart:

A Donut chart was inserted for the patient split by age group. It uses the Legend and Values wells.



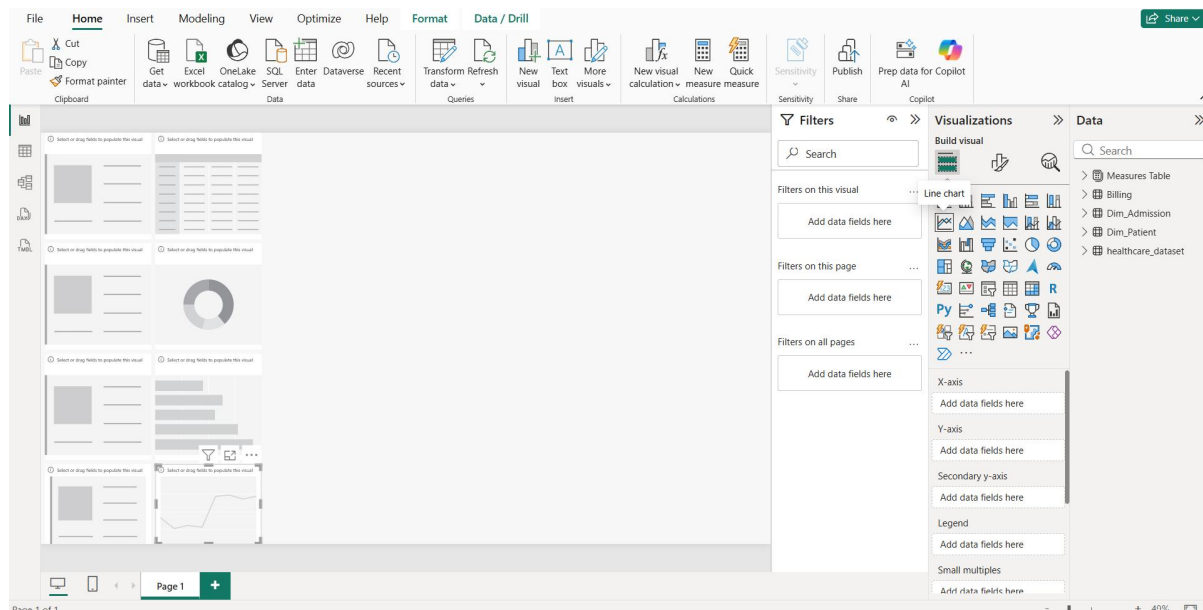
5.4 Adding the Stacked Bar Chart:

A Stacked bar chart was added for patients by medical condition and admission type, with wells for Y-axis, X-axis and Legend.



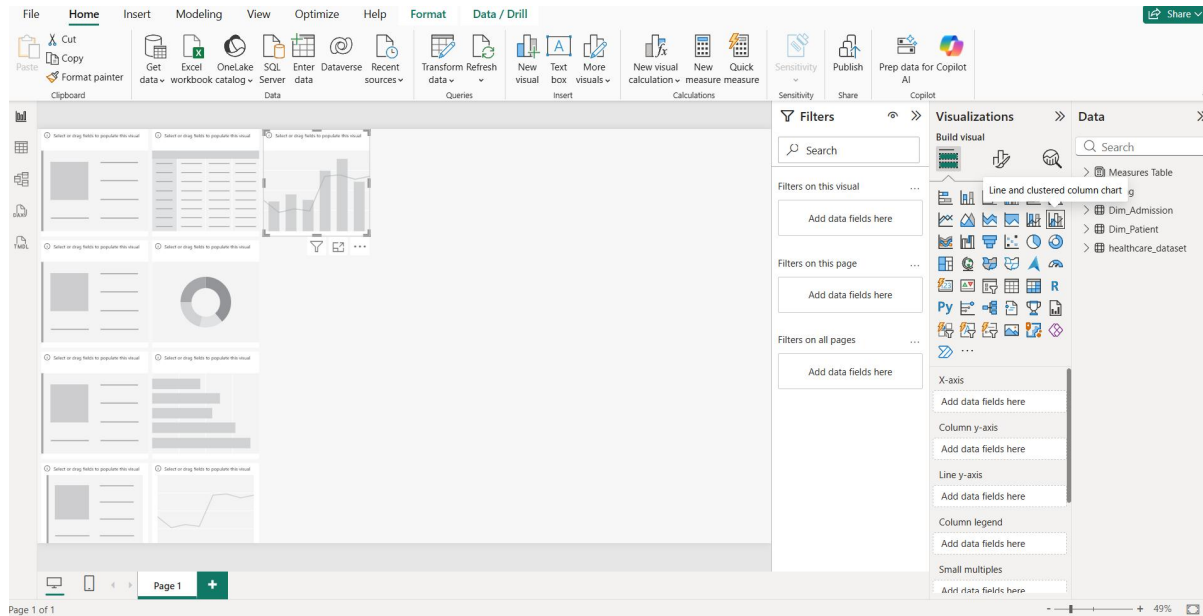
5.5 Adding the Line Chart:

A Line chart was added for the admission trend. Its wells are X-axis, Y-axis, Secondary y-axis and Legend.



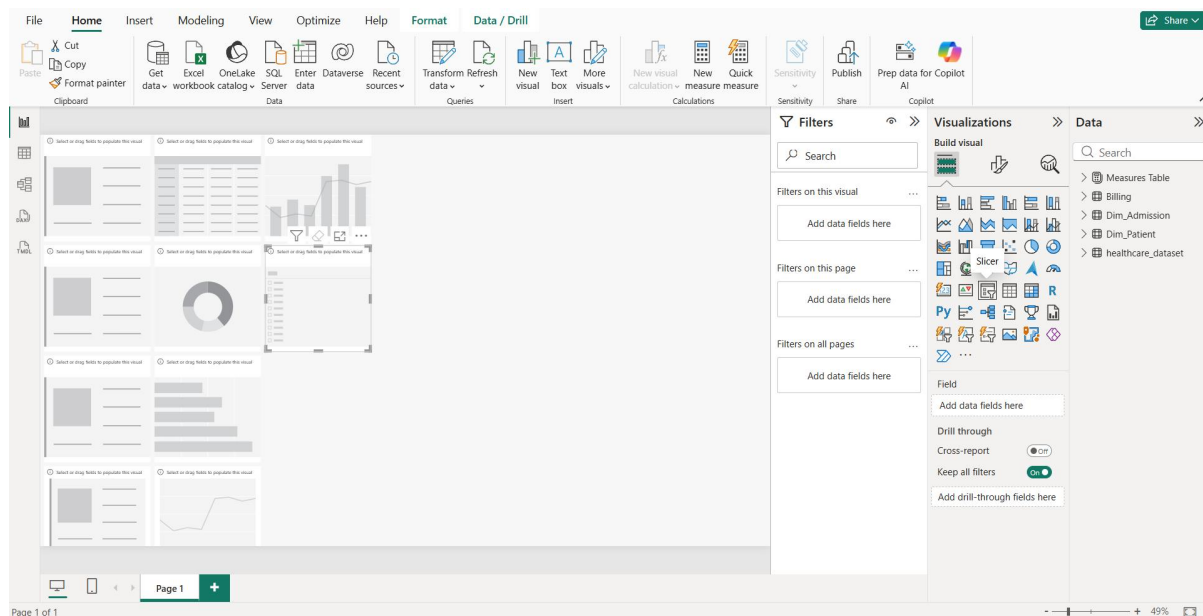
5.6 Adding the Line and Clustered Column Chart:

A Line and clustered column chart was added in the top-right area for billing by insurance provider. Only the column part is used for this visual.



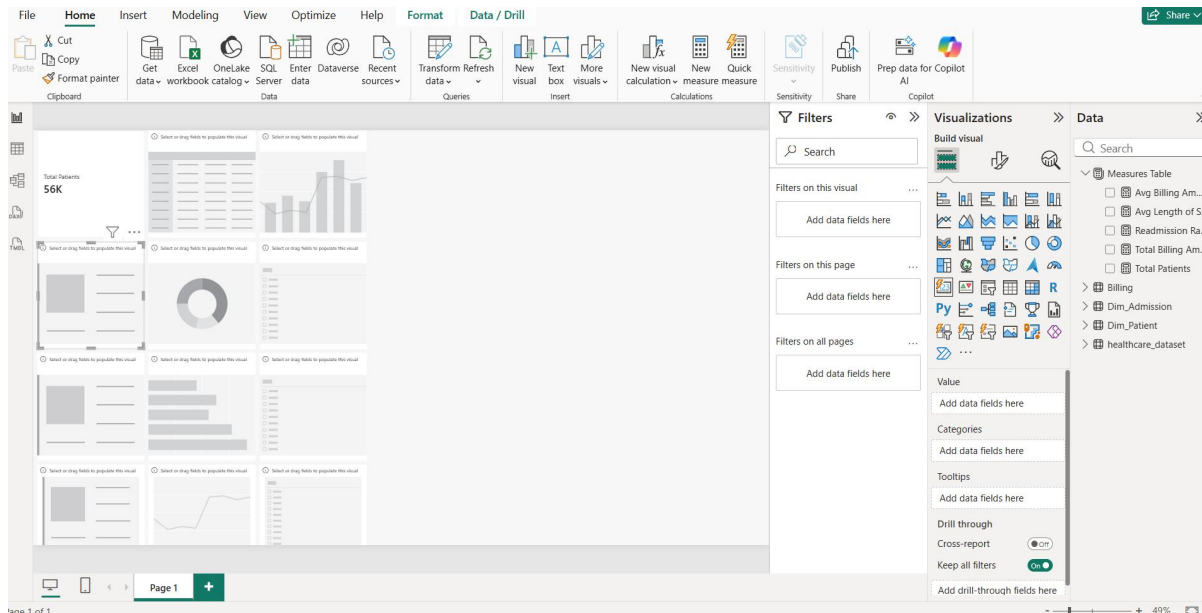
5.7 Adding the Slicers:

Slicer visuals were added for filtering. Each slicer has a single Field well; one slicer was created for Insurance Provider, one for Admission Type and one for Date of Admission.

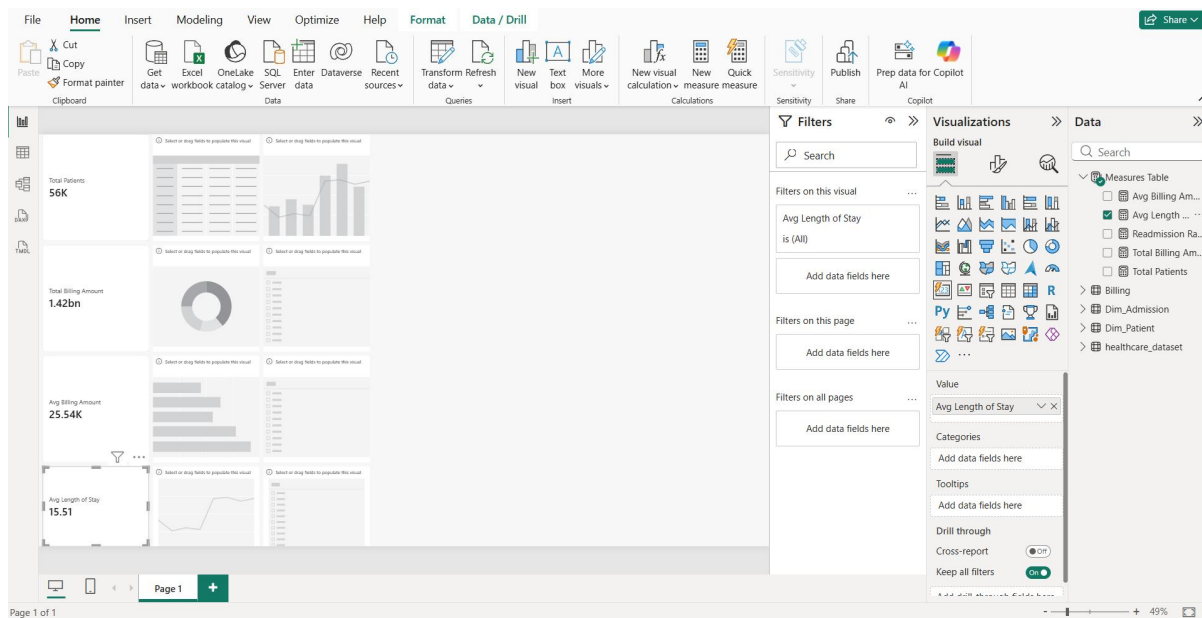


5.8 Binding Data to the KPI Cards:

The measures were then dragged from the Measures Table into the Value well of each card. The first card now shows Total Patients, while the next card is selected and waiting for its measure.

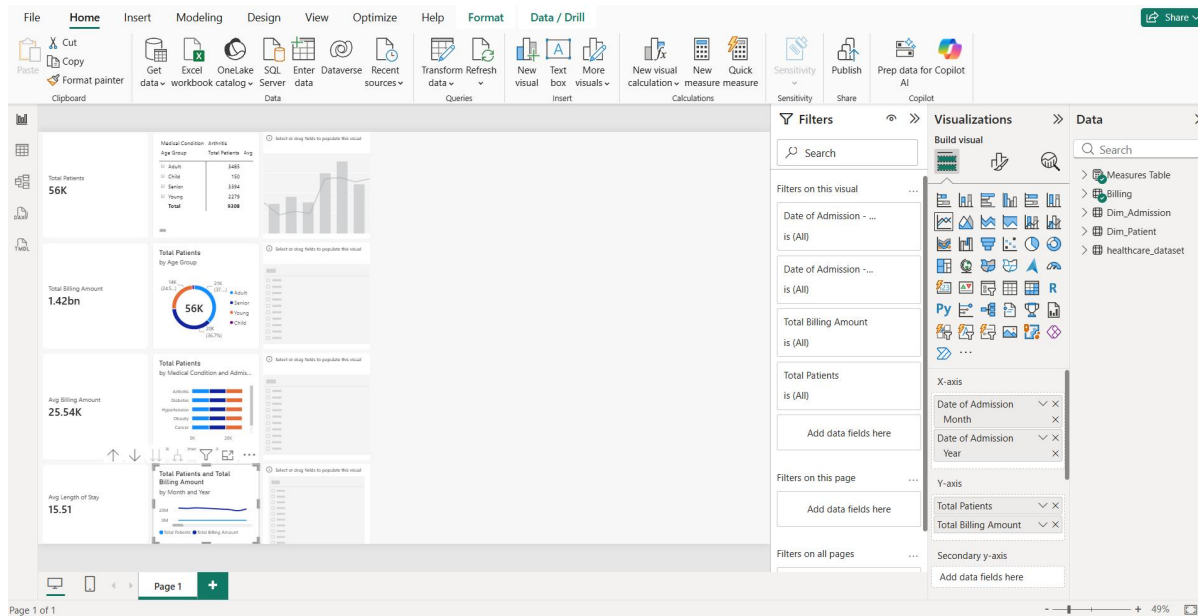


Once all four cards were bound, they show Total Patients, Total Billing Amount, Avg Billing Amount and Avg Length of Stay. The screenshot shows the last card, Avg Length of Stay, assigned to its Value well.



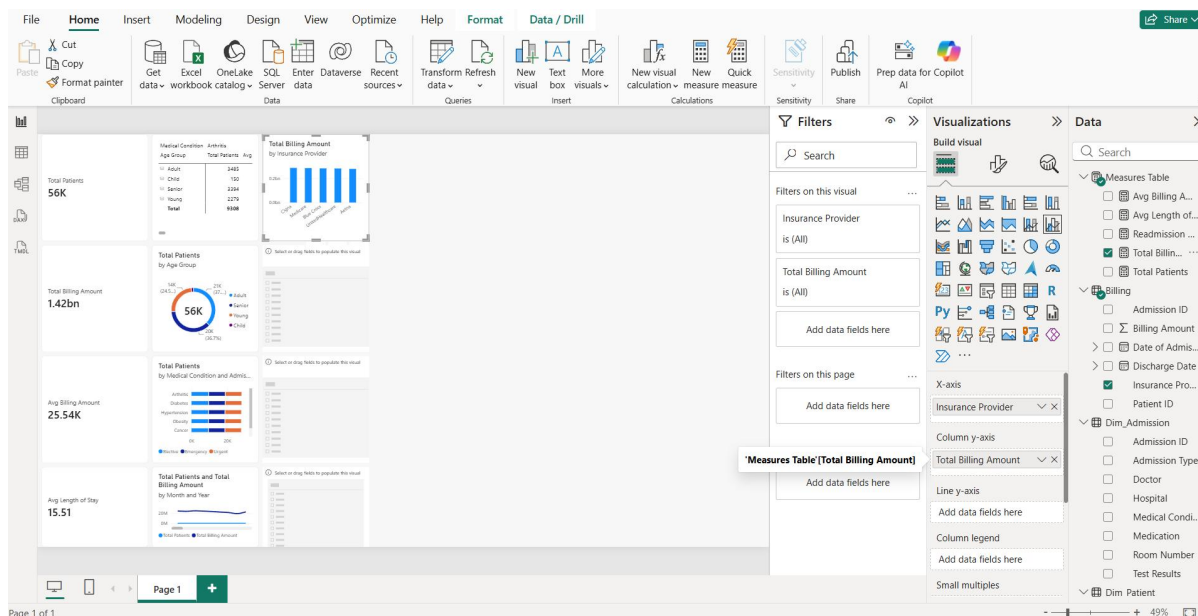
5.9 Building the Trend Line Chart:

For the line chart, Date of Admission (Month and Year) was placed on the X-axis, and Total Patients and Total Billing Amount on the Y-axis, giving a month-by-year trend of admissions and billing.



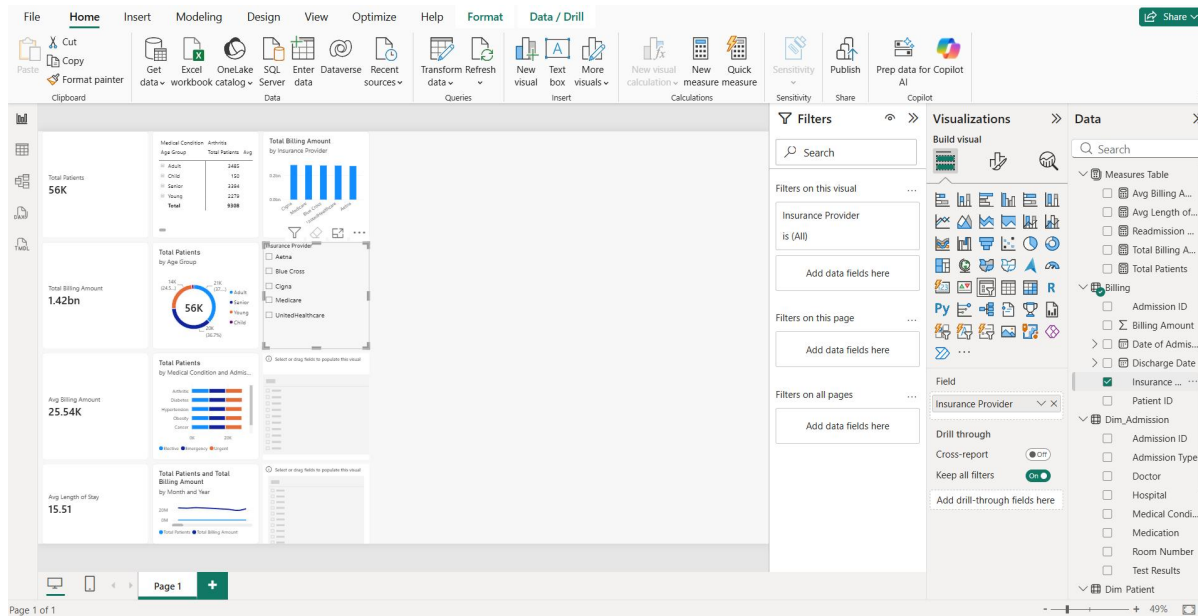
5.10 Building the Billing by Insurance Provider Chart:

For the column chart, Insurance Provider was placed on the X-axis and the Total Billing Amount measure on the Column y-axis, showing billing by insurance provider.



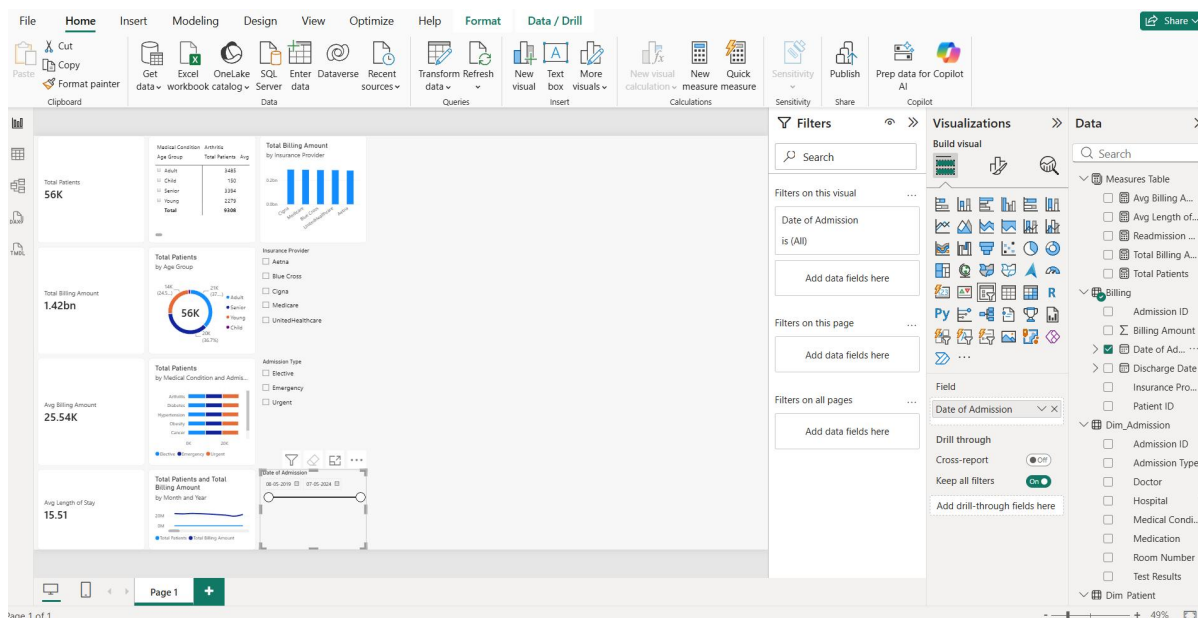
5.11 Configuring the Insurance Provider Slicer:

The Insurance Provider field was added to the first slicer. It lists Aetna, Blue Cross, Cigna, Medicare and UnitedHealthcare as checkbox options. The Admission Type slicer was set up in the same way using the Admission Type field.



5.12 Configuring the Date of Admission Slicer:

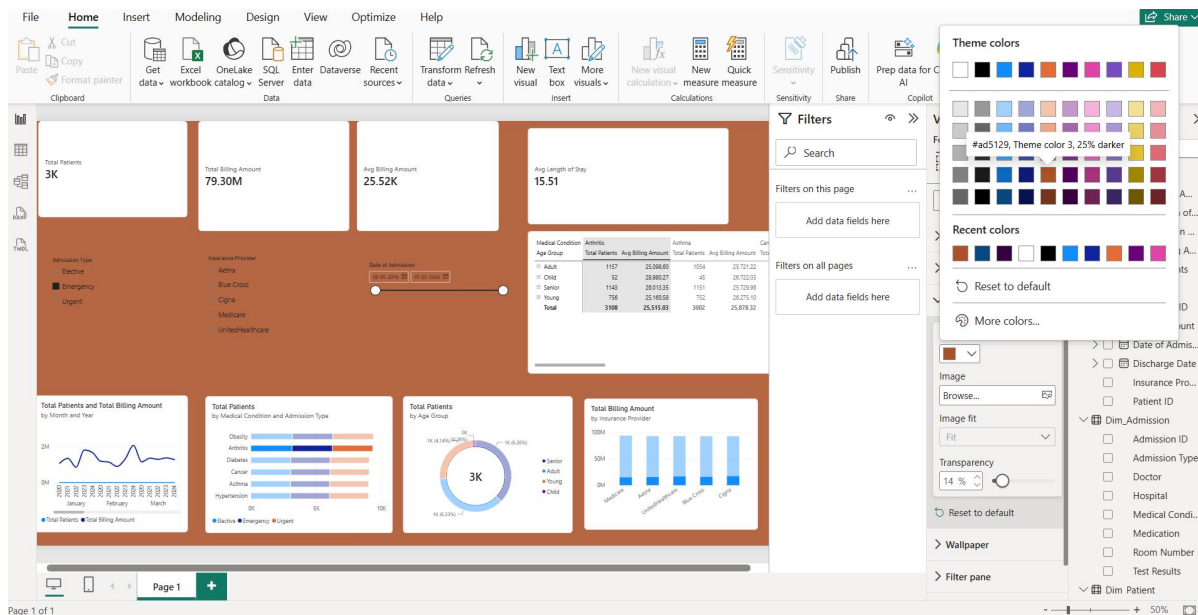
The Date of Admission field was added to the last slicer, which appears as a date-range slider. It lets users narrow every visual to a chosen admission period.



6. Final Dashboard:

The completed dashboard brings together every planned element: four KPI cards at the top, the Admission Type, Insurance Provider and Date of Admission slicers, a matrix by age group and medical condition, and four charts along the bottom (admission trend, patients by condition and admission type, patients by age group, and billing by insurance provider).

For the final formatting, the page background was changed in the Format pane, under Canvas background, to the orange-brown theme colour #ad5129 (Theme color 3, 25% darker), with the transparency set to 14%. The visuals sit on white card backgrounds so they stay readable against the coloured canvas.



All visuals respond to the Insurance Provider, Admission Type, and Date of Admission slicers, giving a single interactive view for exploring patient volume, cost and length of stay across the hospital network. In the screenshot the Emergency admission type is selected, which is why the KPI cards show filtered values (3K patients, 79.30M total billing) instead of the full-dataset figures.

7. Conclusion:

This task walked through the full Power BI workflow: designing a star-schema data model, isolating calculations in a dedicated Measure Table, writing and validating five DAX measures (including a readmission-rate calculation using SUMMARIZE and FILTER), planning a visual layout against business questions, and assembling those visuals one by one into a single interactive dashboard with cross-filtering slicers.